

Zhi WANG

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RESEARCH INTERESTS

My research focuses on the development of **general-purpose decision-making agents**, spanning: i) Reinforcement Learning, ii) Post-Training and Alignment of Foundation Models, iii) Vision-Language-Action Models, and iv) Embodied AI. My long-term goal is to bridge foundation models and embodied intelligence, enabling intelligent agents to perceive, reason, decide, and act autonomously in complex real-world environments.

EXPERIENCE

2023.07–Now	Associate Professor, <i>Nanjing University</i>
2019.11–2023.06	Associate/Assistant Research Fellow, <i>Nanjing University</i>
2026	Visiting Scholar, <i>Nanyang Technological University</i> , Singapore (Host: Prof. Dacheng Tao)
2022	Visiting Scholar, <i>Chinese Academy of Sciences</i> (Host: Prof. Yuanheng Zhu, Dongbin Zhao)
2019	Visiting Scholar, <i>University of New South Wales</i> , Australia (Host: Prof. Daoyi Dong)

EDUCATION

2015.09–2019.10	Ph.D.	<i>City University of Hong Kong</i>	(Supervisor: Prof. Han-Xiong Li)
2011.09–2015.08	B.E.	<i>Nanjing University</i>	(Supervisor: Prof. Chunlin Chen)

SELECTED PUBLICATIONS

Journal Articles

TPAMI 2025	Zichuan Liu, Yuanyang Zhu, Zhi Wang* , Yang Gao, and Chunlin Chen. <i>MIXRTs: Toward Interpretable Multi-Agent Reinforcement Learning Via Mixing Recurrent Soft Decision Trees</i> . (IEEE Transactions on Pattern Analysis and Machine Intelligence)
TNNLS 2024	Donghan Xie, Zhi Wang* , Chunlin Chen, and Daoyi Dong. <i>Depthwise convolution for multi-agent communication with enhanced mean-field approximation</i> . (IEEE Transactions on Neural Networks and Learning Systems)
TNNLS 2024	Jinmei Liu, Zhi Wang* , Chunlin Chen, and Daoyi Dong. <i>Efficient Bayesian policy reuse with a scalable observation model in deep reinforcement learning</i> .
TNNLS 2023	Zhi Wang , Chunlin Chen, and Daoyi Dong. <i>Instance weighted incremental evolution strategies for reinforcement learning in dynamic environments</i> .
TCYB 2023	Zhi Wang , Chunlin Chen, and Daoyi Dong. <i>A Dirichlet Process Mixture of Robust Task Models for Scalable Lifelong Reinforcement Learning</i> . (IEEE Transactions on Cybernetics)

TNNLS 2022	Zhi Wang , Chunlin Chen, and Daoyi Dong. <i>Lifelong incremental reinforcement learning with online Bayesian inference</i> .
TSYS 2021	Zhi Wang and Han-Xiong Li. <i>Dissimilarity analysis-based multimode modeling for complex distributed parameter systems</i> . (IEEE Transactions on Systems, Man, and Cybernetics: Systems)
TNNLS 2020	Zhi Wang , Han-Xiong Li, and Chunlin Chen. <i>Incremental reinforcement learning in continuous spaces with policy relaxation and importance weighting</i> .
TCYB 2020	Zhi Wang , Han-Xiong Li, and Chunlin Chen. <i>Reinforcement learning-based optimal sensor placement for spatiotemporal modeling</i> .
TMECH 2019	Zhi Wang , Chunlin Chen, Han-Xiong Li, Daoyi Dong, and Tzyh-Jong Tarn. <i>Incremental reinforcement learning with prioritized sweeping for dynamic environments</i> . (IEEE-ASME Transactions on Mechatronics)
TSYS 2019	Zhi Wang and Han-Xiong Li. <i>Incremental spatiotemporal learning for online modeling of distributed parameter systems</i> .

Conference Papers

ICML 2026 (Oral)	Haoran Zhang, Yafu Li, Zhi Wang , Zhilin Wang, Shunkai Zhang, Xiaoye Qu, Yu Cheng. <i>Characterizing, Evaluating, and Optimizing Complex Reasoning</i> . (International Conference on Machine Learning)
ICML 2026	Wenhao Wu, Zhentao Tang, Yafu Li, Shixiong Kai, Mingxuan Yuan, Zhenhong Sun, Chunlin Chen, Zhi Wang* . <i>From Conflict to Consensus: Boosting Medical Reasoning via Multi-Round Agentic RAG</i> .
ICLR 2026	Zican Hu, Shilin Zhang, Yafu Li, Jianhao Yan, Xuyang Hu, Leyang Cui, Xiaoye Qu, Chunlin Chen, Yu Cheng, Zhi Wang* . <i>Diversity-Incentivized Exploration for Versatile Reasoning</i> . (International Conference on Learning Representations)
ICLR 2026	Runzhe Zhan, Yafu Li, Zhi Wang , Xiaoye Qu, Dongrui Liu, Jing Shao, Derek F. Wong, Yu Cheng. <i>ExGRPO: Learning to Reason from Experience</i> .
ICLR 2026	Jinmei Liu, Fuhong Liu, Zhenhong Sun, Jianye Hao, Bo Wang, Huaxiong Li, Daoyi Dong, Chunlin Chen, Zhi Wang* . <i>Scalable In-Context Q-Learning</i> .
NeurIPS 2025	Shilin Zhang, Zican Hu, Wenhao Wu, Xinyi Xie, Jianxiang Tang, Chunlin Chen, Daoyi Dong, Yu Cheng, Zhenhong Sun, and Zhi Wang* . <i>Text-to-Decision Agent: Offline Meta-Reinforcement Learning from Natural Language Supervision</i> . (Annual Conference on Neural Information Processing Systems)
NeurIPS 2025	Wenhao Wu, Fuhong Liu, Haoru Li, Zican Hu, Daoyi Dong, Chunlin Chen, and Zhi Wang* . <i>Mixture-of-Experts Meets In-Context Reinforcement Learning</i> .
NeurIPS 2025	Jianhao Yan, Yafu Li, Zican Hu, Zhi Wang , Ganqu Cui, Xiaoye Qu, Yu Cheng, and Yue Zhang. <i>Learning to Reason under Off-Policy Guidance</i> .
ICML 2025	Zican Hu, Wei Liu, Xiaoye Qu, Xiangyu Yue, Chunlin Chen, Zhi Wang* , and Yu Cheng. <i>Divide and Conquer: Grounding LLMs as Efficient Decision-Making Agents via Offline Hierarchical Reinforcement Learning</i> .
NeurIPS 2024	Zhi Wang , Li Zhang, Wenhao Wu, Yuanheng Zhu, Dongbin Zhao, and Chunlin Chen. <i>Meta-DT: Offline Meta-RL as Conditional Sequence Modeling with World Model Disentanglement</i> .

ICLR 2024	Zican Hu, Zongzhang Zhang, Huaxiong Li, Chunlin Chen, Hongyu Ding, and Zhi Wang *. <i>Attention-Guided Contrastive Role Representations for Multi-Agent Reinforcement Learning.</i>
AAAI 2019	Zhi Wang , Wei Bi, Yan Wang, and Xiaojiang Liu. <i>Better fine-tuning via instance weighting for text classification.</i> (AAAI Conference on Artificial Intelligence)

Note: *indicates the corresponding author

SOCIAL SERVICES

- *Program Chair*, IEEE Conference on Cybernetics, 2026
- *Area Chair*, NeurIPS 2026 / AAMAS 2026
- *Guest Editor*, IEEE Transactions on Cybernetics
- *Associate Editor*, IEEE SMC 2026, 2021-2023 / IEEE ICNSC 2020
- *Reviewer*, IEEE TPAMI/TNNLS/TMECH/TCYB/TSYS, ICLR/NIPS/ICML/CVPR/AAAI

HONORS AND AWARDS

- Youth Science and Technology Talent Support Project, Jiangsu Province, 2024
- Zijin Scholar, Nanjing University, 2024
- Youth Science and Technology Award, Jiangsu Automation Society, 2023
- Doctor of Innovation and Entrepreneurship, Jiangsu Province, 2020
- Outstanding Academic Performance Award, City University of Hong Kong, 2018
- Research Tuition Scholarship, City University of Hong Kong, 2018

TEACHING

- <Deep Reinforcement Learning>, mainly for postgraduates, 2019–Now
- <Digital Circuits>, for undergraduates, 2022–Now
- <Introduction to Automation>, for undergraduates, 2019–2022

INVITED TALKS

- Toward efficient reinforcement fine-tuning for AI agents, CCF-AI 2025
- Toward efficient reinforcement fine-tuning of LLMs, Chinese Academy of Sciences, 2025
- Lifelong reinforcement learning for dynamic environments, the 4th Distributed Artificial Intelligence (DAI) Conference, 2022
- Incremental reinforcement learning for dynamic environments, University of New South Wales, 2019